

Hug start, then the fluid equation

The circle is the thing. The cube is only the grid. The hug is the smooth start. After time zero the gate does not act.

Equation

$d_t u + (u \cdot \text{grad}) u = -\text{grad } p + \nu * \text{laplacian } u$, and $\text{div } u = 0$.

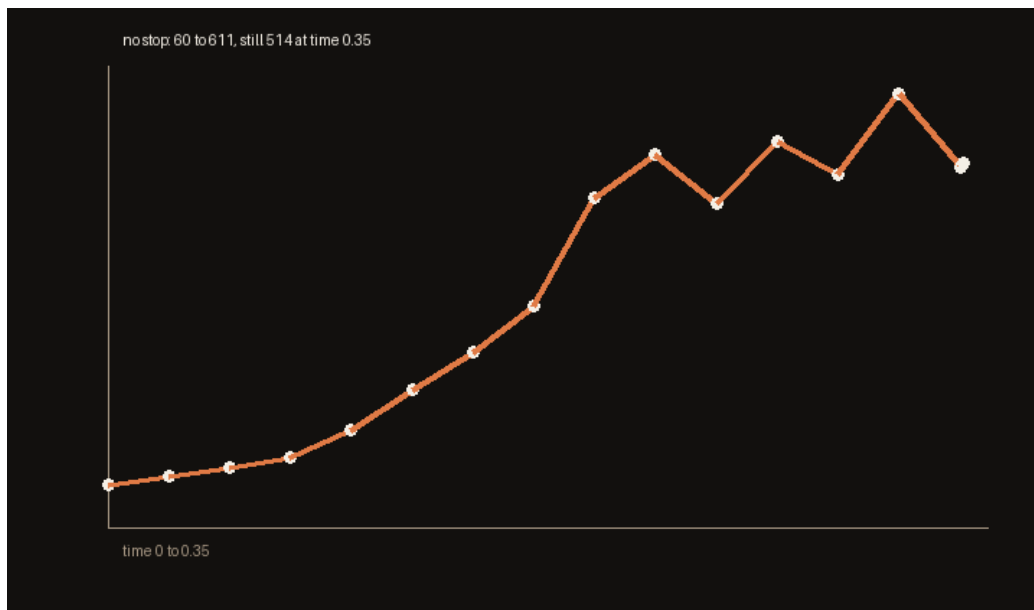
No body force. The gate multiplies the stream function before the velocity is made. It is not a wall after time zero.

Runs

Run	What happened
Ring, ν 0.01	Fades. Speed 0.06 at time 40.
Tighter ring, 64 cubed	Fades. Peak 5.47 to 0.86 by time 2.
Pulled tube, ν 0.002	Speeds up, 3.74 to 5.22, then fades to 1.46.
Same tube, 64 cubed	Climb bigger, 3.60 to 6.16, then below the start.
Two opposite tubes	Never beat the start. 2.98 to 2.19.
Smoothing vs steepening	Smoothing smaller at the fastest point. Speed still fades.
Pressure at that point	Switches sign. Not a steady brake.
Spin ratio, pulled tube	Falls, 21 to 11. Also falls with no smoothing.
Tube in a stretch	Ratio rises 30.6 to 32.2, then falls to 21.3.
Matched stretch, no cap	Biggest spin 60 to 611, still 514 at time 0.35.
Matched stretch, 64 cubed	201 at time 0.12. Not a coarse-grid effect.

The climb that did not settle

A tight tube in a stretch matched to the spin. No cap was left on.



Bound

If the time integral of the biggest spin stays finite, the flow stays smooth up to that time. That is conditional.

From the climbing start alone, the biggest spin is 60.2. The crude earliest blowup is time 0.0166. The run passed that and was still finite at time 0.35. The fastest growth was 16 percent of the square rate.

On the rising steps the growth looked closer to a multiple of the spin than to the spin squared. A multiple of the spin does not explode in finite time. The multiple did not hold the whole way. The spin also dipped.

Open: a bound on the biggest spin that is known before the run and does not run out. These runs do not give that bound.